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A fluconazole population pharmacokinetics study to improve target attainment in critically ill patients

Action needed

You need to make revisions to your submission

Your reviewers have provided feedback on your submission. There are revisions you need to make before your submission can progress. We've also emailed erwin.dreesen@kuleuven.be with more details.

We recommend submitting your revisions by 07 Aug 2025.

Feedback from the Editor

▼ Read editor's comments

Dear Dr Dreesen,

Your manuscript, "A fluconazole population pharmacokinetics study to improve target attainment in critically ill patients", has now been assessed.

We invite you to revise your paper, carefully addressing the comments from the reviewers and the editor. Please ensure the results are accurately reported, any overstated conclusions are rewritten and the limitations of the work fully explained. When your revision is ready, please submit the updated manuscript and a point-by-point response. This will help us move to a swift decision.

We recommend submitting all revisions within the mentioned deadline.

If you need more time, please contact us and include your submission ID.

Kind regards,

Birgit Willinger

Editor

Infection

[Support contact: saritha.hemanth@springernature.com](mailto:saritha.hemanth@springernature.com)

Submission ID: 309c3b72-0e51-43aa-972c-a19890672f43

Feedback from the Reviewer

▼ Read reviewer's comment

Reviewer 1

This article is a population pharmacokinetics (popPK) study of fluconazole aimed at improving target attainment in critically ill patients. Overall, the research design, methodology, and presentation of results are quite comprehensive. However, there are still some aspects that may need further discussion or clarification. The following are some potential issues and areas to consider :

- 1.The study should indicate in the abstract section how to adjust the fluconazole dosage for ICU patients on CRRT, and for patients not on CRRT weighing more than 60 kg.
- 2.The study integrates data from eight different studies, which increases the sample size but may introduce heterogeneity. Differences in patient selection, sampling times, and analytical methods across these studies could affect the homogeneity of the data and the reliability of the results. For example, the recording of CRRT (continuous renal replacement therapy) types and parameters may not be consistent across studies. This should be discussed in the limitations part of the study.
- 3.Covariate Handling: While stepwise covariate modeling was used to select significant covariates, the interactions between these covariates (e.g., between CRRT status and eGFR) may not have been fully considered. Additionally, the handling of missing data (e.g., through multiple imputation) is reasonable but may introduce additional uncertainty.
- 4.Threshold Setting for Target Attainment (PTA): Although the study set $fAUC_{0-24}/MIC = 100$ as the target, the MICs of different strains may vary significantly in clinical practice, which could affect the practical significance of the target attainment rate. Moreover, for strains with higher MICs (e.g., $MIC = 4$ mg/L), the proposed dosing regimen may not be sufficient to achieve the target.
- 5.Feasibility of Dose Adjustment: The study proposes an optimized dosing regimen based on patient total body weight and CRRT status, but frequent dose adjustments may be challenging in actual clinical practice, especially in the ICU setting where patients' conditions and physiological parameters can change rapidly.
- 6.The authors stated in the Results section: "The standard loading dose did not result in clinically acceptable PTA of $\geq 90\%$ when total body weight exceeded 65 kg and 30 kg, in patients off and on CRRT, respectively." However, in the Abstract, they mentioned: "With standard dosing, only CRRT status and total body weight were clinically relevant, as PTA dropped below 90% for all patients on CRRT, and for patients not on CRRT weighing more than 60 kg." The authors should clarify whether it is 60 kg or 65 kg, and ensure consistency between the two sections.

Attachment(s):

- [Download Reviewer 1 attachment 1](#)

Reviewer 2

Dear authors,

Please display our main conclusion already in the abstract!

You applied the CDK EPI equation for estimation of the GFR (eGFR). The eGFR, however, is well established in patients with stable renal impairment, but generally unreliable parameter in critically ill because of changes in renal function during severe illness. Please discuss this problem in the limitations section.

You have focussed on pharmacokinetic data derived from previous studies and, therefore, you have not assessed toxicity or adverse effects in your present study. Only, the risk of trough concentrations above 80 mg/l was calculated. Unlike the label, you do not recommend a reduction of the maintenance dose for patients with a GFR < 30 ml/min. Fluconazole is relatively well tolerated, but it might cause hepatotoxicity as well as QT prolongation at ECG bearing the risk of life-threatening arrhythmias. Therefore, please discuss safety aspects in the light of the relatively high loading doses you recommend!

In the legend of figure S12 you mention a “vertical dashed green line” that “illustrates the start of fluconazole infusion”. I cannot see this line. Please clarify!

How to submit your revision

The original submitting author must upload a point-by-point response to the comments as a PDF file. This must include a description of any additional experiments that were carried out and a detailed rebuttal of any criticisms or requested revisions that you disagreed with.

Any files (including the manuscript) that have changed based on the comments will also need to be uploaded again. Do not include tracked changes in your manuscript file. If you need to upload a marked-up version of the revised manuscript with the changes highlighted you can upload it on the related file section.

Please note the original submitting author may be different from the corresponding author.

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Progress so far

Progress so far

1. Submission received - complete

- 2. Technical check - complete
- 3. Editorial assignment - complete
- 4. With editor - complete
- 5. Peer review - action needed

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Your submission

Your submission

Title
A fluconazole population pharmacokinetics study to improve target attainment in critically ill patients

Type
Research

Journal
Infection

Submission ID
309c3b72-0e51-43aa-972c-a19890672f43

Submission version
v.1.0

Manuscript File
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Submission history

1. **Submission received**

Submission status	Date
Submission received	03 May 2025
2. **Technical check**

Submission status	Date
Submission is under technical check	03 May 2025
Submission passed technical check	05 May 2025
3. **Editorial assignment**

Submission status	Date
Ready for editorial assignment	05 May 2025
4. **With editor**

Submission status	Date
Editor assigned	05 May 2025
5. **Peer review**

Submission status	Date
First reviewer(s) invited	10 May 2025
Reviewer(s) accepted	11 May 2025

Reviewer report(s) received 24 Jun 2025

Revision requested 10 Jul 2025

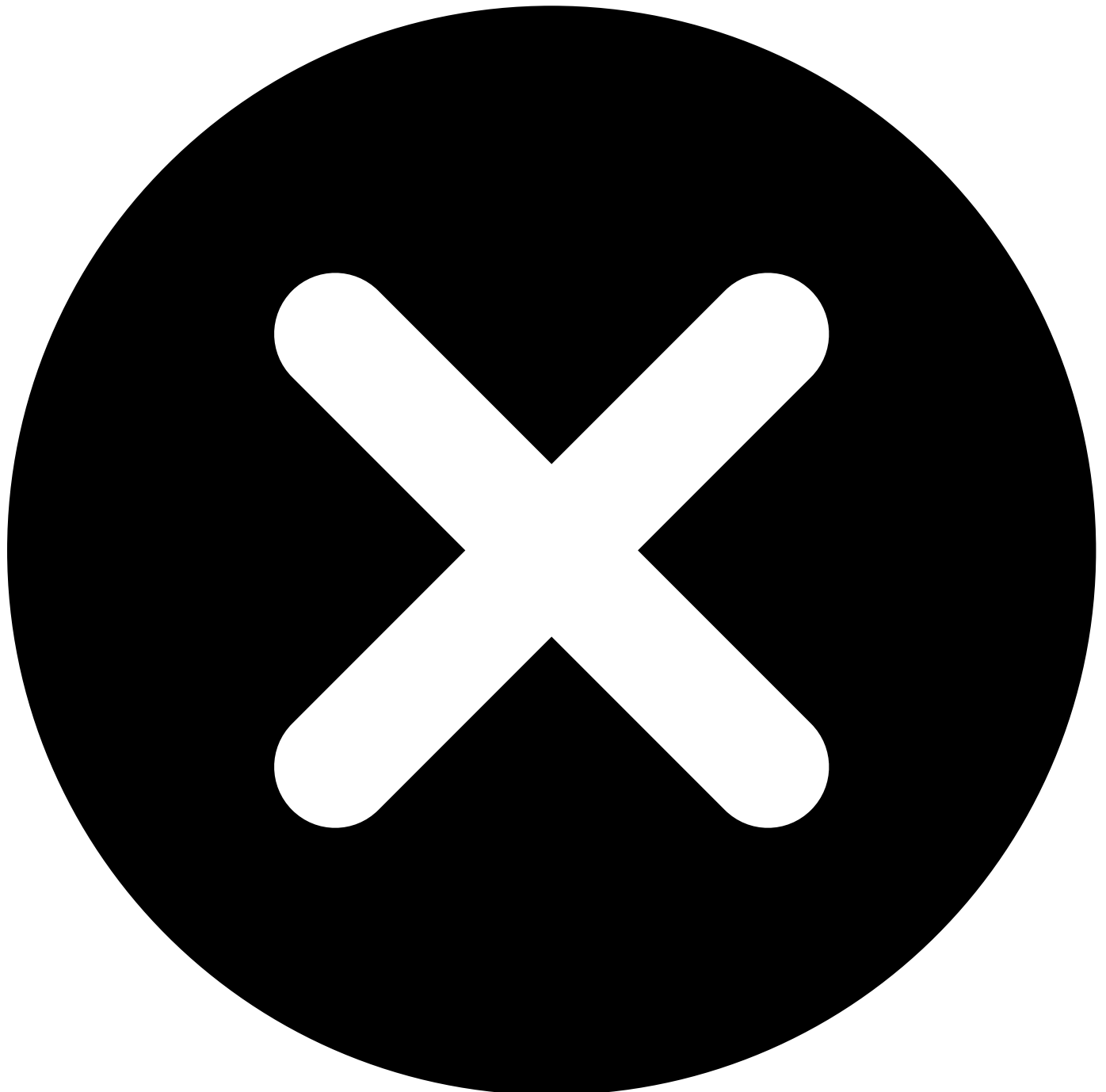
Need help?

If you have any questions about this submission, you can [email the Editorial Office](mailto:saritha.hemanth@springernature.com) (saritha.hemanth@springernature.com).

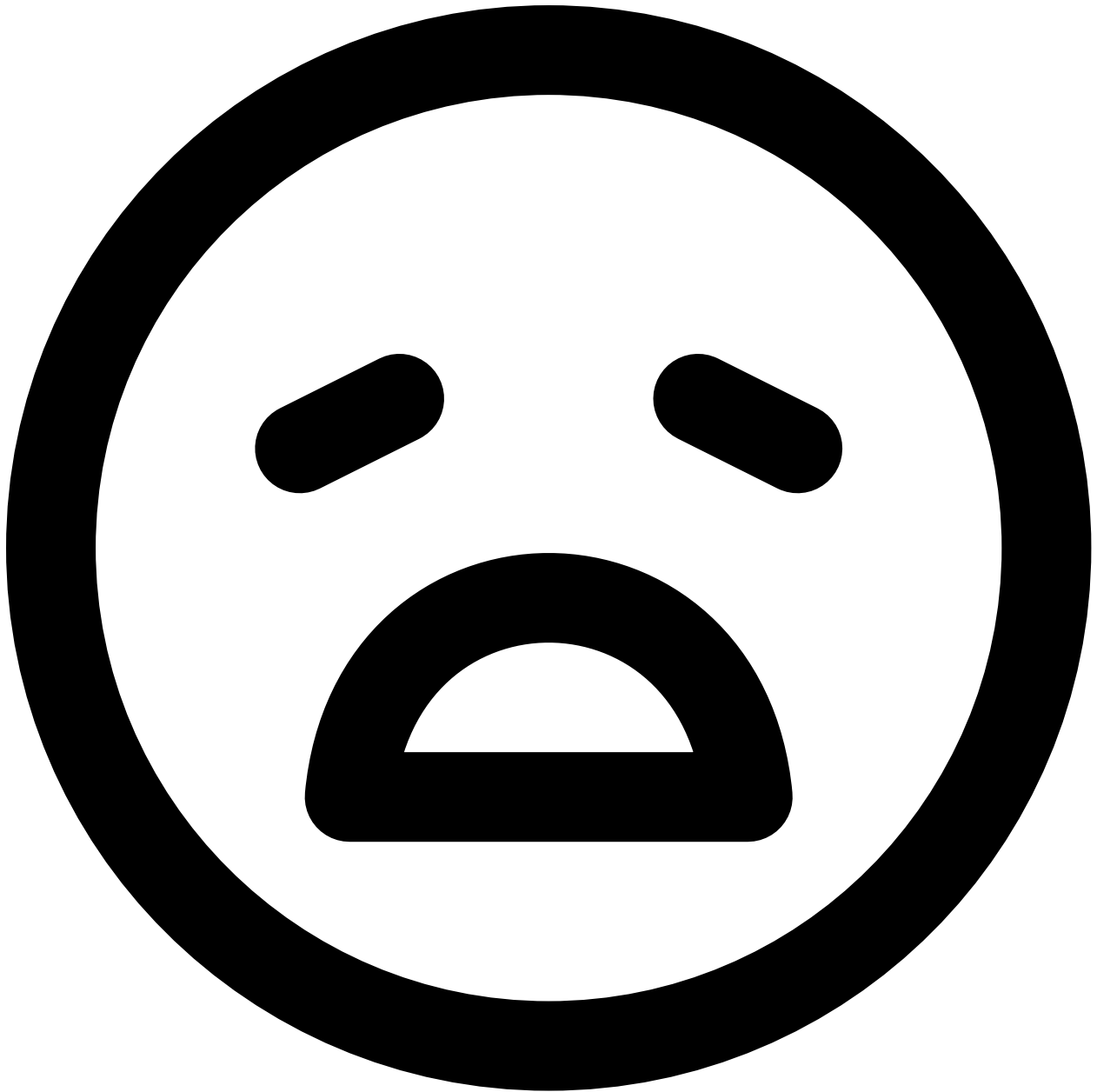
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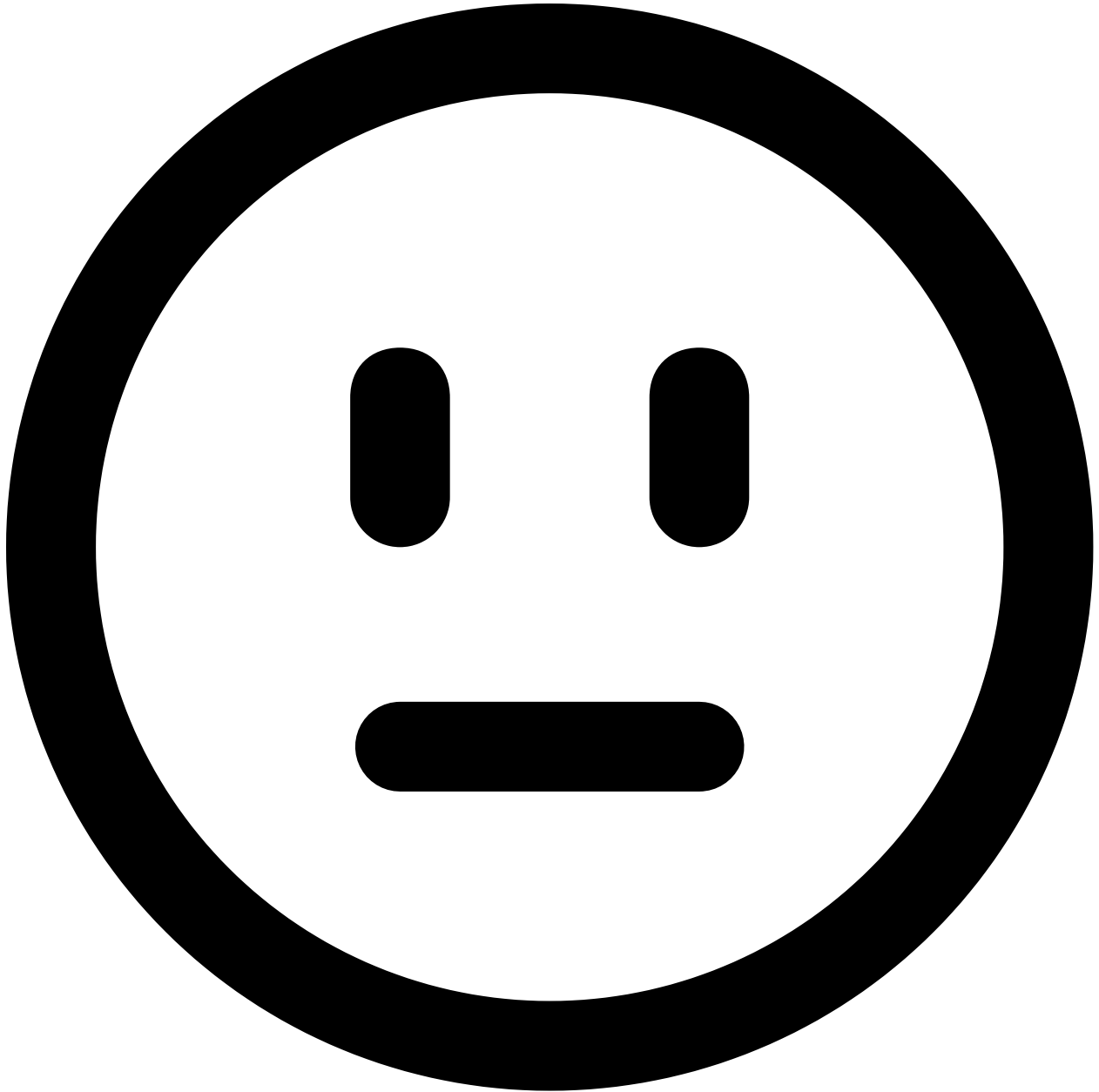
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